



COLLEGE OF ENGINEERING SCIENCE

& TECHNOLOGY



COMPUTER AND SOFTWARE DEPARTMENT

FACULTY OF ENGINEERING

Computer and Software Engineering Department (CSE)

Prof. Tamer M. Nassef

Introduction

Computer and Software Engineering is concerned with the design of computer systems. Computer Engineering spans a broad range of activities, from the design and modeling of devices used in the construction of computer systems to the configuration of large systems and networks of computers, including both hardware and software.

The application of computer engineering ranges from the design of highly-specialized, dedicated processors to general-purpose systems, and the physical size of computing systems range from microprocessors to supercomputers. Upon completing the computer and software engineering program the student will also gain a solid foundation in the basic sciences, mathematics, electrical theory, and computer hardware and software. While the program includes a strong core curriculum, individual interests and specific career goals may be accommodated through electives in technical specialties. During the senior year, students are permitted to select technical electives either in one or more specialization.

Computer and Software Engineering graduates find employment in industry government, and educational institutions. Opportunities also exist for self-employed computer consultants. Specific jobs for which the computer engineers are prepared include hardware design, design automation, VLSI design, test design, fault tolerant design, software engineering, digital communications, artificial systems development, computer networking and computer architecture. For this reason, Misr University for Science and Technology is offering this program to fill a very important gap in the job market in Egypt.

The general character of this Computer and Software Engineering curriculum is designed with the aim of providing the student with fundamental training in Computer and Software Engineering disciplines equal to those offered by American Universities in about 142 credit hours, and is extended further towards enabling him/her to acquire sufficient Computer and Software Engineering education within the range of 40 more credit hours to end up with a total of 182 credit hours.

To insure the right merging of the Computer and Software Engineering concepts within the different areas of this engineering profession, the student is advised to go into little more depth in one of the sub-disciplines of Computer and Software Engineering through various combinations of electives.

Mission Statement:

The department provides facilities to students for having leadership roles in Computer & Software Engineering. These facilities include high-quality education to support and enhance the profession of Computer & Software Engineering through its education, research and service activities. Students will be taught to address problems of programming on solid technical foundations while taking advantage of advanced technologies. The graduates will be prepared for the pursuit of advanced learning in Computer & Software Engineering and related fields, as well as for the practice at the highest professional levels.

Objectives

The objective of this program is to prepare engineers at the entry professional level to conceive, plan, design, direct, estimate the cost and safely manage the problem solving process by computer. The program is designed to turn out future Computer & Software engineers who will understand and accept the moral, ethical, legal and professional obligations for protecting the public welfare. Also, the graduates will acquire the kind of critical thinking and creativity needed for successful development based on gaining the leadership and management skills. This will create effective teams and communication routines.
Accordingly, within the vision and the mission of the department, the objectives are:

- The department will provide high quality education that prepares undergraduate and graduate students for a life-long learning experience.
- The Computer & Software Engineering program provides its graduates with skills necessary to design a solution for a given problem using engineering components (Hardware & Software) to meet desired goals. Graduates are expected to think creatively, synthesize information, and to come up with practical solutions for close- and open-ended practical design problems.
- Graduates should be able to function effectively individually, as well as in multi-disciplinary teams with ability to communicate effectively both orally and in writing.
- Graduates are expected to develop broad knowledge of current technologies important to Computer & Software Engineering. They are expected to gain proficiency in the use of modern engineering techniques and tools, with emphasis on the role of computers and information technology in engineering.
- Graduates are equipped with broad education in Computer Engineering as well as Software Engineering for other technical and non-technical disciplines.
    - Appreciate and understand the impact of Computer & Software Engineering and technology in a global and social context including environmental issues.
    - Develop solid understanding of professional and ethical responsibilities.
    - Demonstrate the ability to engage in lifelong learning in their profession as well as contemporary issues pertaining to their profession and society in which they live and work.

Special point's in the program

To insure the right merging of the Computer & Software Engineering concepts within the different areas of the Engineering profession and also to maintain the features of the Computer Engineering program, the student is advised to go into little more depth in one of the sub-disciplines in database design, operating systems, programming languages and algorithms, or to develop his interest through various combinations of electives.
Although the program was recently started, it had gained excellent reputation. The graduates of this program are ready to work in different fields of design, management as well as decision making. Also, they are capable of learning and improving on the job thus filling a very important gap in the job market in Egypt as well as in other Middle East countries. Also, there is always immediate and wide spread increase in demand for Computer & Software Engineers from both government and private sectors, giving rise to many job opportunities. This can be manifested by the continuous demand for computer engineers by Agencies, Consulting firms,

General communications companies, Research organizations, as well as Sales engineering companies.

البرنامج الدراسي لقسم هندسة الحاسبات والبرمجيات

(Computer and Software Engineering Department)

يدرس الطالب (120) ساعة معتمدة كمتطلبات تخصص موزعة على النحو التالي :

◄(108) ساعة معتمدة متطلبات إجبارية موزعة على النحو التالي :

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
ECE 201	Electronics I	3	2	2	1	5	PHY 102
ECE 202	Electronics II	3	2	2	1	5	ECE 201
ECE 203	Logic Design	3	2	2	1	5	MATH 101
ECE 211	Circuits I	3	2	2	1	5	MATH 102 PHY 102
ECE 212	Circuits II	3	2	2	1	5	ECE 211
ECE 232	Electromagnetics I	3	2	2	-	4	PHY 201
ECE 321	Measurements I	3	2	2	1	5	ECE 212
ECE 341	Signal Analysis	3	2	2	-	4	MATH 202
ECE 342	Communications I	3	2	2	1	5	ECE 341
ECE 402	Digital Signal Processing I	3	2	2	-	4	ECE 341
ECE 423	Measurements II	3	2	2	1	5	ECE 321
ECE 451	Systems Engineering	3	2	2	-	4	ECE 341
ECE 552	Control I	3	2	2	1	5	ECE 451
ECE 595	Speech and Image Processing	3	2	2	-	4	ECE 402
MATH 202	Linear Algebra and Differential Equations	3	2	2	-	4	MATH 201
MATH 302	Numerical Analysis	3	2	2	-	4	MATH 202
PHY 201	Physics III	3	2	-	3	5	PHY 102
ME 300 C	Fundaments of Thermo fluids	3	2	-	3	5	PHY 102
CSE 301	Data Base Systems	3	2	-	3	5	CSE 308
CSE 302	Software Engineering	3	2	-	3	5	CSE 308
CSE 303	Assembly Language	3	2	-	3	5	CSE 307

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE 307	Microprocessors	3	2	-	3	5	ECE 203
CSE 308	Programming and Algorithms	3	2	-	3	5	COMP 101
CSE 401	Operating Systems I	3	2	2	-	4	CSE 308
CSE 402	Computer Organization	3	2	2	-	4	CSE 307
CSE 404	Real Time Computer Systems	3	2	2	-	4	CSE 307
CSE 405	Operating Systems II	3	2	2	-	4	CSE 401
CSE 491	Computer Interfacing	3	2	-	3	5	CSE 307
CSE 500	Computer Networks I	3	2	0	3	5	ECE 342
CSE 503	Artificial Intelligence	3	2	-	3	5	CSE 405
CSE 506	Data Structures	3	2	2	-	4	CSE 308
CSE 586	Computer Systems Programming	3	2	-	3	5	CSE 404
CSE 588	Pattern Recognition	3	2	2	-	4	ECE 402
CSE 589	Neural Networks	3	2	2	-	4	ECE 402
CSE 590	Expert Systems	3	2	2	0	4	CSE 503
CSE 595	Object-Oriented Programming.	3	2	2	0	4	CSE 506

◄كما يدرس الطالب (12) ساعة معتمدة متطلبات تخصص اختيارية من بين أحدي المجموعات الآتية :

CSE Concentration Area (1): Computer Networks

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE 555	Network Programming and Routing	3	2	2	0	4	CSE 500
CSE 556	Information Networks.	3	2	2	0	4	CSE 500
CSE 557	Internet Protocols	3	2	2	0	4	CSE 500
CSE 558	High Speed Networks	3	2	2	0	4	CSE 500

CSE Concentration Area (2): Systems and Data Security

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE 565	Introduction to Cryptography	3	2	2	0	4	CSE 308
CSE 566	Data Security	3	2	2	0	4	CSE 308
CSE 567	Computer Network Security.	3	2	2	0	4	CSE 500
CSE 568	Access Control and Physical Security	3	2	2	0	4	CSE 402
CSE 505	Computer Organization II	3	2	2	0	4	CSE 402
CSE 509	Computer Aided Design	3	2	2	0	4	CSE 307

CSE Concentration Area (3):Computerized Control Systems

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE 575	Instrumentation and Process Control	3	2	2	0	4	ECE 451
CSE 576	PLCs & Microcontroller Based Systems	3	2	2	0	4	CSE 307
CSE 577	Microprocessors Based Systems	3	2	2	0	4	CSE 307
CSE 578	Fuzzy Control Systems and MATLAB	3	2	2	0	4	ECE 552
CSE 581	Computer Manufacturing Technology	3	2	2	0	4	CSE 307

CSE Concentration Area (4):Design of Smart Systems

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE 585	Algorithms Design & Analysis.	3	2	2	0	4	CSE 308
CSE 596	Artificial Intelligence II	3	2	2	0	4	CSE 503
CSE 597	Compiler Design	3	2	2	0	4	CSE 405
CSE 505	Computer Organization II	3	2	2	0	4	CSE 402
CSE 509	Computer Aided Design	3	2	2	0	4	CSE 307
CSE 510	Modeling & Simulation	3	2	2	0	4	CSE 404

◄كما يقضى الطالب 45 يوما تدريبا ميدانيا فى مواقع صناعية وهندسية على النحو التالى:

Course Code	Course Title	Credit Hours	Contact Hours	Number of Days	Pre-Req.
CSETR 201	Field Training II	---	75	15	GETR 101
CSETR 301	Field Training III	---	75	15	CSETR 201
CSETR 401	Field Training IV	---	75	15	CSETR 301

الخطة الدراسيه لقسم هندسةالحاسبات والبرمجيات

First Level

Semester (1)*(Common for All Departments)*

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
ENGL 101	English I	3	2	2	-	4	-
MATH 101	Calculus I	3	2	2	-	4	-
PHY 101	Physics I	3	2	-	3	5	-
GE 101	Introduction to Engineering	2	2	-	-	2	-
GE 102	Manufacturing Technology	2	1	-	3	4	-
GE 103	Engineering Graphics	2	1	-	3	4	-
GE 104	Engineering Mechanics I	3	2	2		4	-
Total		**18**	**12**	**6**	**9**	**27**	

Semester (2) (Common for All Departments*)*

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
ENGL 102	English II	3	2	2	-	4	ENGL 101
MATH 102	Calculus II	3	2	2	-	4	MATH 101
CHEM 101	Chemistry	3	2	2	1	5	-
PHY 102	Physics II	3	2	-	3	5	-
COMP101	Introduction to Computers	3	2	-	3	5	-
GE105	Engineering Mechanics II	3	2	2	-	4	GE 104
Total		**18**	**12**	**8**	**7**	**27**	

Summer Semester I (Common for All Departments*)*

Course Code	Course Title	Credit Hours	Contact Hours	Number of Days	Pre-Req.
GETR 101	Field Training I	---	75	15	25 Credits

Second Level

Semester (3)

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
ECE 201	Electronics I	3	2	2	1	5	PHY 102
ECE 203	Logic Design	3	2	2	1	5	MATH 101
ECE 211	Circuits I	3	2	2	1	5	MATH 102 PHY 102
ENGL 201	English III	3	2	2		4	ENGL102
MATH 201	Vector Algebra and Calculus III	3	2	2		4	MATH 102
PHY 201	Physics III	3	2	-	3	5	PHY 102
Total		**18**	**12**	**10**	**6**	**28**	

Semester (4)

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE 307	Microprocessors	3	2	-	3	5	ECE 203
ECE 202	Electronics II	3	2	2	1	5	ECE 201
ECE 212	Circuits II	3	2	2	1	5	ECE 211
CSE 308	Programming and Algorithms	3	2	-	3	5	COMP 101
MATH 202	Linear Algebra and Differential Equations	3	2	2		4	MATH 201
ME 300 C	Fundaments of Thermo fluids	3	2	-	3	5	PHY 102
Total		**18**	**12**	**6**	**11**	**29**	

Summer Semester II

Course Code	Course Title	Credit-hours.	Contact hours	Number of Days	Pre-Req.
CSETR 201	Field Training II	---	75	15	GETR 101

Third Level

Semester (5)

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
ARAB 101	Arabic Language	3	3	-	-	3	-
CSE 506	Data Structures	3	2	2	-	4	CSE 308
ECE 232	Electromagnetics I	3	2	2	-	4	PHY 201
ECE 341	Signal Analysis	3	2	2	-	4	MATH 202
CSE 491	Computer Interfacing	3	2	-	3	5	CSE 307
MATH 301	Engineering Probability and Statistics	3	2	2	-	4	MATH 102
Total		**18**	**13**	**8**	**3**	**24**	

Semester (6)

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
MATH 302	Numerical Analysis	3	2	2	-	4	MATH 202
GE 205	Engineering Economics	2	2	-	-	2	-
ECE 342	Communications I	3	2	2	1	5	ECE 341
ECE 451	Systems Engineering	3	2	2	-	4	ECE 341
ECE 402	Digital Signal Processing I	3	2	2	-	4	ECE 341
CSE 301	Data Base Systems	3	2	-	3	5	CSE 308
CSE 302	Software Engineering	3	2	-	3	5	CSE 308
Total		**20**	**14**	**8**	**7**	**29**	

Summer Semester (III)

Course Code	Course Title	Credit Hours	Contact hours	Number of Days	Pre-Req.
CSETR 301	Field Training III	---	75	15	CSETR 201

Fourth Level

Semester (7)

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE xxx	Elective Course	3	2	2	-	4	-
ECE 552	Control I	3	2	2	1	5	ECE 451
ECE 321	Measurements I	3	2	2	1	5	ECE 212
CSE 303	Assembly Language	3	2	-	3	5	CSE 307
HUMN102	Scientific Thinking	3	3	-	-	3	-
CSE 401	Operating Systems I	3	2	2	-	4	CSE 308
Total		**18**	**13**	**8**	**5**	**26**	

Semester (8)

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE 404	Real Time Computer Systems	3	2	2	-	4	CSE 307
CSE XXX	Elective Course	3	2	-	3	5	-
CSE 500	Computer Networks I	3	2	-	3	5	ECE 342
CSE 405	Operating Systems II	3	2	2	-	4	CSE 401
ECE 423	Measurements II	3	2	2	1	5	ECE 321
CSE 588	Pattern Recognition	3	2	2	-	4	ECE402
Total		**18**	**12**	**8**	**7**	**27**	

Summer Semester (VI)

Course Code	Course Title	Credit Hours	Contact Hours	Number of Days	Pre-Req.
CSETR 401	Field Training IV	---	75	15	CSETR 301

Fifth Level

Semester (9)

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
ECE 595	Speech and Image Processing	3	2	2	-	4	ECE 402
CSE 589	Neural Networks	3	2	2	-	4	ECE 402
CSE 503	Artificial Intelligence	3	2	-	3	5	CSE 405
CSE 590	Expert Systems	3	2	2	0	4	CSE 503
CSE xxx	Elective Course	3	2	2	-	4	-
CSE 599A	Senior CSE Project I	3	1	-	6	7	134 Cr Hr
Total		**18**	**11**	**8**	**9**	**28**	

Semester (10)

Course Code	Course Title	Credit Hours	Lec.	Tut.	Lab.	Contact Hours	Pre-Req.
CSE 402	Computer Organization	3	2	2	-	4	CSE 307
CSE 595	OBJECT ORIENTED PROGRAMMING	3	2	2	-	4	CSE 506
CSE 586	Computer Systems Programming	3	2	-	3	5	CSE 404
HUMN xxx	University Elective	3	3	-	-	3	-
CSE XXX	Elective Course	3	2	2	-	4	-
CSE 599B	Senior CSE Project II	3	1	-	6	7	134 Cr Hr
Total		**18**	**12**	**6**	**9**	**27**	

Lec. : *Lecture Hours*

Tut. : *TutorialHours*

Lab. :*Laboratory Hours*

Pre-Req. :*Prerequisites*

Course Descriptions

CSE 301 DATA BASE SYSTEMS 3(2-0-3):

Fundamentals of database, advanced database structures, Database description, Relational database, Distributed database, Database management, Database security, Synchronous access, Database performance evaluation database compression.

Prerequisite: CSE 308

CSE 302 SOFTWARE ENGINEERING 3(2-0-3):

The course covers software design and development topics beyond introductory programming, introducing additional software design and development methods, Alternative operating system and programming environment, Software design and development methods enabling students to undertake larger and more complex programs for computer and electrical engineering applications.

Prerequisite: CSE 308

CSE 303 ASSEMBLY LANGUAGE PROGRAMMING 3(2-0-3):

Introduction to the microprocessor, Addressing modes, Data transfer instructions, Arithmetic and logic instructions, Program control instructions, Programming of the microprocessor, 8085/8086/8088, Hardware specifications, Memory interface, Interrupts, direct memory access and DMA controlled I/O, The family of arithmetic coprocessors, PC Assembly, 80186-80486 microprocessors.

Prerequisite: CSE 307

CSE 307 MICROPROCESSORS 3(2-0-3):

Operation and applications of microprocessors including system level organization, analysis of specific processors and software and hardware interface design, Microprocessor development systems including in-circuit emulation, programming using higher-level languages, implementation of application algorithms.

Prerequisite: ECE 203

CSE 308 PROGRAMMING AND ALGORITHMS I 3(2-0-3):

Efficient design of algorithms, Complexity analysis and its limitation in basic problems, interfacing and compatibility problems, Applications with C++ language, Applications with C#., Concepts of Object Orientation, Introduction to Unified Modeling Language (UML) Design.

Prerequisite: COMP 101

CSE 401 OPERATING SYSTEMS 3(2-2-0):

Basic functions-OS life cycle, Memory management, Virtual memory, Central OS management, Task scheduling Terminal management, File management - I/O systems, Digital computers interconnection structures, Computer components, Computer function interconnections structure and bus system, Computer instruction architecture, Central processing unit (CPU) design, Input/output and communication, Distributed system structure-distributed file systems- distributed coordination- Protection- Security- case study: Windows NT, UNIX systems, Linux systems.

Prerequisite: CSE 308

CSE 402 COMPUTER ORGANIZATION 3(2-2-0)

Digital computers interconnection structures, Computer components, Computer function interconnections structure and bus system, Computer instruction architecture, Central processing unit (CPU) design, input/output and communication.

Prerequisite: CSE 307

CSE 404 REAL TIME COMPUTER SYSTEMS 3(2-2-0):

Introduction, Operation requirements (soft and hardware, Real time modeling, Real time design, Real time operating systems. Data base entry and processing in real time-applications.

Prerequisite: CSE 307

CSE 405 OPERATING SYSTEMS II 3(2-2-0)

Network operation, programmer interface, distributed system architecture, distributed file systems, distributed coordination, Protection, Security, case study, Windows NT, UNIX systems, Linux systems.

Prerequisite: CSE 401

CSE 491 COMPUTER INTERFACING AND PERIPHERALS 3(2-0-3):

Main interfacing units, direct memory access synchronous and asynchronous computer communications, Parallel and series communications peripheral types and its connections, Channel programming, Synchronization and operation control. Case study using C-language

Prerequisite: CSE 307

CSE 500COMPUTER NETWORKS 3(2-0-3):

Introduction, ISO seven layers model, Computer communications, hierarchy and protocols, Network planning and design, Routing algorithms, network planning and design, network management, Local area and distributed networks, MAN and WAN networks. Prerequisite: ECE 342.

CSE 503 ARTIFICIAL INTELLIGENCE 3(2-0-3):

Languages for programming, knowledge and reasoning, problem solving, informed search and exploration, learning methods of artificial intelligence, communicating, perceiving and acting. Prerequisite: CSE 405.

CSE 505 COMPUTER ORGANIZATION II 3(2-2-0)

Instruction set design, Memory organization and management, Single stream control; Object oriented processing, Single I-stream parallelism, Parallelism by message passing, Shared resources systems, Protection and security.
Prerequisite: CSE 402

CSE 506 DATA STRUCTURES 3(2-2-0):

Data transmission, Transmission media, Long distance communication, Hardware addressing and frame type identification, Physical topology and interface hardware, Data encoding, Data communication interface.
Prerequisite: CSE 308.

CSE 507 DIGITAL DESIGN (COMPUTER CIRCUITS) 3(2-0-3):

An introduction to programmable logic devices, Synchronous binary counter design. Decision making with binary counters, Output forming logic design, Programming PLD circuits with CUPL, Design examples using PLDs, ROM-based controller architecture, Microprocessor architecture, The MC68HCII Microcontrollers, Address bus decoding and decoding and logic design.
Prerequisite: ECE 203

CSE 509 COMPUTER AIDED DESIGN 3(2-2-0)

Design methodology for digital systems using a modern hardware description language (HDL). Learn algorithm, architecture, and implementation aspects of arithmetic processing elements such as adders, multipliers, and dividers. Study architecture designs of Complex and Reduced Instruction Set Computer (CISC, RISC). Floating –point arithmetic processors. Emphasize parallelism and pipelining techniques in digital system design. The use of a simulation/synthesis environment of the HDL as well as other CAD tools such as a layout tool and a tester, evaluate, implement as hardware, and test such digital systems. Prerequisite: CSE 307

CSE 510 Modeling & Simulation 3(2-2-0)

The course is intended to provide you with a reasonable grounding in how to make use of modeling techniques in order to evaluate the dynamic behavior of real systems. This includes choosing an appropriate modeling formalism, constructing the model, verifying the model and solving the model. The main emphasis of the lectures will be on the different modeling formalisms and model construction. However, through the many examples which will be presented in the lectures and in the notes you will also gain experience of deciding the right level of abstraction, verification and solution. You are strongly encouraged to take advantage of the example models available from these web pages and to experiment for yourselves.
Prerequisite: CSE 404

CSE 555 NETWORK PROGRAMMING AND ROUTING 3(2-2-0):
Time and frequency signals, Analog and digital signals, High level data link control, Ethernet operations, collision and broadcast domains, Routed and IP routing protocols, sub-netting mechanismsPrerequisite: CSE 500

CSE 556 INFORMATION NETWORKS 3(2-2-0):
User Datagram Protocol (UDP), UDP applications, Transmission Control Protocol (TCP), TCP/IP applications, File Transfer Protocol (FTP), Simple Mail Transfer Protocol (SMTP), Network File System (NTS)
Prerequisite: CSE 500.

CSE 557 INTERNET PROTOCOLS3(2-2-0):
Internet protocols, IP fragmentation, Address resolution protocol, RARP packet format, IP helpers, Internet Control Message Protocol, Internet Group Management Protocol, IP routing and multicasting, Dynamic routing protocols.
Prerequisite: CSE 500

CSE 558 HIGH SPEED NETWORKS 3(2-2-0):
Integrated Service Digital Networks (ISDN), Frame relay, Broadband ISDN, Digital subscriber DSL, Asynchronous Transfer Mode (ATM)
Prerequisite: CSE 500

CSE 565 INTRODUCTION TO CRYPTOGRAPHY 3(2-2-0):
Basic concepts of cryptography, data security information theory, complexity theory and number theory, cipher techniques for encryption
Prerequisite: CSE 308

CSE 566 DATA SECURITY 3(2-2-0):
Classical encryption techniques, Data Encryption Standards (DES), Advanced Encryption Standards (AES), Triple DES, Blowfish, RC5, Key Distribution, Random Number Generation, Public Key Crypto-mechanisms , The RSA algorithm.
Prerequisite: CSE 308

CSE 567 COMPUTER NETWORK SECURITY 3(2-2-0):
Network Security Services, Security mechanisms, Security Attacks, the OSI security architecture, A model for network security, Authentication requirements and functions, Message authentication Code (MACs), Hash functions, Hash algorithms, Digital signature and authentication protocols, applications in securityPrerequisite: CSE 500

CSE 568 ACCESS CONTROL AND PHYSICAL SECURITY 3(2-2-0):
Access Matrix Model, Access Control Mechanisms, Access Hierarchy, Authorization Lists, Identification, Smart Cards and Tokens, Biometric, Audit, Physical Access Control Techniques, ID card management and alarm monitoring.
Prerequisite: CSE 402

CSE 575 INSTRUMENTATION AND PROCESS CONTROL 3(2-2-0):

Principles of modeling, developing of empirical models, linear regression in modeling, frequency response, Role of process and simulation in modeling.Concepts of PID controller. Feed-forward and feedback control strategy.
Prerequisite: ECE 451.

CSE 576 PLCs AND MICROCONTROLLER BASEDSYSTEMS 3(2-2-0):
Conceptual design of relay logic, flexibility of programmable logic controllers, arithmetic and logic operations with PLC, working with analogue modules, RISC programming with microcontroller, working with digital I/O , A/D on board converters and serial communication with Microcontroller.
Prerequisite: CSE 307

CSE 577 MICROPROCESSOR BASED SYSTEMS 3(2-2-0):
Advanced architecture of microprocessor, pipelining and concurrent processing, multithreading and multitasking, thread priority, processor bench working enhancement, bootstrapping algorithms
Prerequisite: CSE 307

CSE 578 FUZZY CONTROL SYSTEMS AND MATLAB 3(2-2-0):
Difference between crisp logic and fuzzy logic, membership functions, fuzzification and defuzzification techniques, improving response with neural networks (Neuro fuzzy), fuzzy inference system in MATLAB.
Prerequisite: ECE 552

CSE 581 COMPUTER MANUFACTURING TECHNOLOGY 3(2-2-0)
Microprocessor architecture, programming and architecture, computer files and debug, 8086 instruction and segmentation, EXE files and miscellaneous instruction, Linking object files and ASY library, advanced memory and port I/O, string operation, interrupts and I/O, assembly language, memory and I/O routines
Prerequisite: CSE 307

CSE 585 ALGORITHMS DESIGN AND ANALYSIS 3(2-2-0):
Introduction to algorithms and review of data structure; time and space analysis; algorithm design techniques: divide and counter greedy algorithms, dynamic programming, search techniques; NP-complete problems and approximation algorithms. Structured languages
Prerequisite: CSE 308

CSE 586 COMPUTER SYSTEMS PROGRAMMING 3(2-0-3):
Systems software and its relationship to machine architecture design of assemblers, loaders linkers, and microprocessors. Device drivers.Editors, Debuggers.Brief introduction to operating systems and compilers.
Prerequisite: CSE 404

CSE 588 PATTERN RECOGNITION 3(2-2-0):

Introduction to computer vision, Feature Vectors and Classifiers, Supervised Versus Unsupervised Pattern Recognition, classifier based on Baycs Decision Theory, parametric techniques (maximum likelihood estimation, maximum a posterior estimation and estimation maximization), nonparametric techniques (nearest neighborhood), Linear Discriminated Function (Perception Algorithm, Mean Square Error Estimation), nonlinear classifier, feature selection and generation techniques, feature generation, matching, unsupervised learning and clustering Analysis, examples and design using MATLAB.

Prerequisite:ECE 402

CSE 589 NEURAL NETWORKS 3(2-2-0):

Neuron Model and Network Architectures, Multiple layers of neuron. Perception Learning Rule, Supervised Hebbian Learning, Widrow-Hoff Learning, Adaptive Learning neuron – Back-propagation multilayer perception, Associative Learning, Competitive Networks, examples and designs using MATLAB.

Prerequisite:ECE 402

CSE 590 EXPERT SYSTEM 3(2-2-0):

Alternate knowledge Base representation, Characterization of expert systems (statistical methods, structural methods, hybrid methods, perception, and scene analysis), inference engine design and semantic network, predicate Language characteristics, expert systems shell design.

Prerequisite: CSE 503

CSE 595 OBJECT ORIENTED PROGRAMMING 3(2-2-0):

Class design with Universal Modeling Language (UML), concepts of encapsulation, abstraction, and polymorphism, class Inheritance, Practical applications.

Prerequisite: CSE 506

CSE 596 ARTIFICIAL INTELLIGENCE II 3(2-2-0):

Advanced artificial intelligence topics.

Prerequisite: CSE 503

CSE 597 COMPILER DESIGN3(2-2-0):

Various parsing techniques, symbol table handing, Lexical Analyzer, Syntax Analyzer, Semantic Analyzer, Intermediate and Final Code Generation, Code Optimization, Memory Allocation and Implementation Issues, run-time memory management, error handling and code optimization techniques.

Prerequisite: CSE 405.

CSE 599A SENIOR CSE PROJECT I 3(1-0-6):

CSE 599B SENIOR CSE PROJECT II 3(1-0-6):

The student undertakes an independent design project in one of the sub-disciplines of computer and software engineering under the supervision of a faculty advisor. The objective is to provide the student with an opportunity to integrate and apply the knowledge he gained through his course in an actual problem, particularly use of design and construction methodologies, evaluation of alternate solutions, feasibility considerations and detailed system descriptions. Realistic design and management constraints including economic factors and reliability will be imposed. The student must document his study in a technical report and give an oral presentation.

Prerequisite:**Completion of 134 credit hours.**